YITIAN WANG

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EDUCATION

University of California, Riverside, CA *Mar. 2021 - Jun. 2026 (expected)*
Ph.D. in Electrical and Computer Engineering
Awards: MRS 2023 Spring Highly Commended Student Talk Award
Dissertation Completion Fellowship Award

Columbia University in the City of New York, NY *Aug. 2019 - Feb. 2021*
Master of Science, Material Science and Engineering

University of California, Berkeley, CA *Jul. 2017 - Aug. 2017*
Summer Session

Beijing Normal University, CN *Sep. 2015 - Jun. 2019*
Bachelor of Science, Physics
Awards: 'Jingshi' Scholarship (2016 - 2018)

SKILLS

Software / Programming

- Python (advanced), MatLab (advanced), SolidWorks (intermediate)
- GSAS-II (advanced), Mantid (advanced), ImageJ (intermediate)

Experimental Techniques

- FZ Crystal Growth (advanced), PVD (intermediate), CVD (intermediate)
- SEM (advanced), XRD (advanced), INS (advanced), Raman (intermediate), FTIR (intermediate)
- PPMS (advanced), DSC (intermediate), TGA (intermediate)
- Prototype Battery Assembly (advanced), Electrochemical Impedance Spectroscopy (intermediate)

PROJECTS

Thermal transport and ionic mobility in solid electrolytes *Jun. 2021 - Jun. 2025*
University of California *Riverside, CA*

· Grew samples with floating zone method using an image furnace.
· Studied the transport properties of complex oxide ionic conductors.

Inelastic neutron scattering on LLZTO single crystal *Jan. 2022 - Jul. 2023*
Oak Ridge National Laboratory *Oak Ridge, TN*

· Conducted neutron scattering experiments using TAX and ARCS.
· Processed and analyzed the data with Mantid and Python.

Two-channel fitting model for thermal conductivity *Mar. 2022 - Sep. 2023*
University of California *Riverside, CA*

· Developed an expandable fitting model using MatLab.

Advanced separator of lithium-ion battery *Oct. 2019 - Jan. 2021*
Columbia University *New York, NY*

· Made a series of composite battery separators with $\gamma - C_3N_4$ and PVDF.
· Assembled full cell batteries and tested their electrochemical properties.

First-principles calculation of olivine structure *Sep. 2019 - Jan. 2020*
Columbia University *New York, NY*

· Used Quantum ESPRESSO to calculate the structure with hydrogen defects.

Low temperature uniaxial strain device *Sep. 2018 - May 2019*
Beijing Normal University *Beijing, CN*

· Modeled devices in SolidWorks to apply pressure utilizing thermal expansion.
· Tested the strain induced detwinning effect with PPMS.

SELECTED PUBLICATIONS

- Low Thermal Conductivity and Lattice Anharmonicity of NaSICON-type Solid Electrolyte $Na_3Zr_2Si_2PO_{12}$
  - *Tungsten*, 2025 (Accepted)
  - **Y. Wang**, Q. Jia, S. Li, L. Shi, Y. Li, X. Chen
- Glass-like thermal transport in polycrystalline perovskite lithium-ion conductor $Li_{3/8}Sr_{7/16}Hf_{1/4}Ta_{3/4}O_3$
  - *Chemical Communications*, 2025
  - **Y. Wang**, Q. Jia, S. Li, L. Shi, Y. Li, X. Chen
- Origin of intrinsically low thermal conductivity in a garnet-type solid electrolyte: Linking lattice and ionic dynamics with thermal transport
  - *PRX Energy*, 2025
  - **Y. Wang**, Y. Su, J. Carrete, H. Zhang, N. Wu, Y. Li, H. Li, J. He, Y. Xu, S. Guo, Q. Cai, D. L. Abernathy, T. Williams, K. V. Kravchyk, M. V. Kovalenko, G. K. H. Madsen, C. Li, X. Chen
- Thermal properties and lattice anharmonicity of Li-ion conducting garnet solid electrolyte $Li_{6.5}La_3Zr_{1.5}Ta_{0.5}O_{12}$
  - *Journal of Materials Chemistry A*, 2024
  - **Y. Wang**, S. Li, N. Wu, Q. Jia, T. Hoke, L. Shi, Y. Li, X. Chen
- Enhanced magnon thermal transport in yttrium-doped spin ladder compounds $Sr_{14-x}Y_xCu_{24}O_{41}$
  - *Journal of Applied Physics*, 2024
  - S. Li, S. Guo, **Y. Wang**, H. Li, Y. Xu, V. Carta, J. Zhou, X. Chen
- Size-dependent magnon thermal transport in a nanostructured quantum magnet
  - *Cell Reports Physical Science*, 2024
  - S. Guo, H. Li, X. Bai, **Y. Wang**, S. Li, R. E. Dunin-Borkowski, J. Zhou, X. Chen
- Crystal structure and thermoelectric properties of layered van der Waals semimetal $ZrTiSe_4$
  - *Chemistry of Materials*, 2022

  - Y. Xu, Z. Barani, P. Xiao, S. Sudhindra, **Y. Wang**, A. A. Rezaie, V. Carta, K. N. Bozhilov, D. Luong, B. P. T. Fokwa, F. Kargar, A. A. Balandin, X. Chen

- Single crystal growth and electrochemical studies of garnet-type fast Li-ion conductors
  - *Tungsten*, 2022
  - **Y. Wang**, X. Chen

EXTRA-CURRICULAR

GSA Python Coding Contest	Aug. 2020
COMAP Mathematical Contest in Modeling	Jan. 2017
BNU Physics Department Soccer Team	2015 - 2019